

# Note on: *Bond Implied CDS Spread and CDS-Bond Basis*

Richard Zhou, 2008

Reading notes

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## Metadata

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## One-paragraph summary

A short, explicit derivation of the CDS spread implied by a bond's market price, obtained from the no-arbitrage condition on a negative basis trade (buy bond, fund at Libor, buy CDS protection on  $1 - D/(1 - R)$  notional — the amount that makes the combined position default neutral). The main formula (7) expresses the bond-implied CDS as the sum of three weighted contributions: bond coupon, an average-Libor-weighted term, and a bond-discount term, all scaled by risky / risk-free annuities. A consistency theorem proves this CDS spread also satisfies the standard CDS pricing identity (5) *if and only if* the survival probabilities and recovery are consistent with the bond price. The analogous decomposition for the par-asset-swap spread (2) and the resulting three-term decomposition of the CDS-bond basis (9) make the basis structure transparent: an interest-rate-curve-slope term, a default-treatment asymmetry term (always negative), and a bond-discount term (positive for discount bonds, negative for premium bonds). Numerical tables on a 10Y 7%-coupon bond confirm the qualitative predictions.

## 1 Setup and notation

| Paper                         | House                         | Meaning                                                              |
|-------------------------------|-------------------------------|----------------------------------------------------------------------|
| $T_0, T_k$                    | $T_0, T_k$                    | Today; bond coupon / ASW payment dates                               |
| $C$                           | —                             | Bond coupon rate (fixed, semiannual in examples)                     |
| $M$                           | —                             | Bond risk-free price                                                 |
| $1 - D$                       | —                             | Bond market (dirty) price; $D$ is the discount from par              |
| $R$                           | Rec                           | Recovery rate (40% in examples)                                      |
| $W$                           | $W = (1 - D/(1 - R))/(1 - D)$ | CDS-notional-to-bond-price ratio (default-neutral)                   |
| $L_{k-1}$                     | $L_{k-1}$                     | Forward Libor for period $(T_{k-1}, T_k)$ at $T_0$                   |
| $DF(T_k)$                     | $D_{T_0, T_k}$                | Risk-free discount factor                                            |
| $P(\tau > T_k)$               | $Q(T_0, T_k)$                 | Survival probability to $T_k$                                        |
| $\lambda$                     | $\lambda$                     | Hazard rate (constant in single-bond calibration)                    |
| $A$                           | $A$                           | Risk-free annuity                                                    |
| PV01                          | —                             | Risky annuity (survival-weighted)                                    |
| $\overline{PV01}$             | —                             | Weighted-average risky annuity (paper's $\widehat{PV01}$ )           |
| $\overline{L}$                | $\overline{L}$                | DF-weighted average forward Libor (eq. ??)                           |
| $\overline{L}^{\text{Risky}}$ | —                             | Risky-DF-weighted average forward Libor                              |
| Loss                          | —                             | Risky default-leg integral $\sum (DF_{k-1} - DF_k)(P_{k-1} + P_k)/2$ |
| $S^{\text{ASW}}$              | —                             | Par asset swap spread                                                |
| $S^{\text{CDS}}$              | —                             | Bond-implied CDS spread                                              |
| Basis                         | —                             | $S^{\text{CDS}} - S^{\text{ASW}}$ , CDS-bond basis                   |

## 2 Key equations

**Risk-free annuity and DF-weighted Libor:**

$$A = \sum_{k=1}^N \Delta T_{k-1} DF(T_k), \quad \overline{L} = \frac{1}{A} \sum_{k=1}^N L_{k-1} \Delta T_{k-1} DF(T_k). \quad (1)$$

**Par asset swap spread (two equivalent forms):**

$$S^{\text{ASW}} = \frac{M - (1 - D)}{A} = C - \overline{L} - \frac{D}{A}. \quad (2)$$

*Says:* the first form is the textbook “risk-free price minus market price, amortised by the annuity”; always non-negative when  $M \geq 1 - D$ . The second form decomposes the spread into *coupon – avg Libor – amortised discount*, which the paper exploits for the basis decomposition. Note:  $-D/A$  is positive when  $D > 0$  (discount bond).

**Default-neutral CDS notional:**

$$W = \frac{1 - D/(1 - R)}{1 - D}. \quad (3)$$

*Says:* buying  $W \cdot (1 - D)$  of CDS protection makes the investor's loss on default equal exactly  $1 - D$ , recovering the purchase price. For a discount bond ( $D > 0$ ),  $W < 1$  (less CDS than notional); for a premium bond,  $W > 1$ .

**Risky annuity and Loss leg:**

$$PV01 = \sum_{k=1}^N \Delta T_{k-1} DF(T_k) P(\tau > T_k), \quad \text{Loss} = \sum_{k=1}^N (DF(T_{k-1}) - DF(T_k)) \frac{P(\tau > T_{k-1}) + P(\tau > T_k)}{2}. \quad (4)$$

**Standard CDS pricing (consistency target):**

$$S^{\text{CDS}} = \frac{(1 - R) \text{Loss}}{PV01}. \quad (5)$$

*Says:* the textbook CDS-fair-spread formula.

**Survival-recovery consistency with bond price:**

$$1 - D = C \cdot PV01 + R \cdot \text{Loss} + P(\tau > T_N) \cdot DF(T_N). \quad (6)$$

*Says:* a defaultable bond decomposes into the risky-discounted coupon stream, the recovery on the default leg, and the survival- weighted final notional. If  $\lambda$  (or a term structure of  $\lambda$ ) is calibrated so that this identity holds at the observed bond price, the model is self-consistent.

**Bond-implied CDS spread (explicit form):**

$$S^{\text{CDS}} = \frac{C \cdot PV01}{W \cdot PV01^{\text{Risky}}} - \frac{\bar{L}^{\text{Risky}} \cdot PV01}{W \cdot PV01^{\text{Risky}}} + \frac{D}{W \cdot PV01^{\text{Risky}}} \left( 1 - \frac{R}{1 - D} \cdot \frac{\text{Loss}}{PV01^{\text{Risky}}} \right), \quad (7)$$

which under (6) simplifies to (paper's eq. 7)

$$S^{\text{CDS}} = \frac{C \cdot PV01 - \bar{L}^{\text{Risky}} \cdot PV01^{\text{Risky}}}{PV01^{\text{Risky}}} + \frac{D}{PV01^{\text{Risky}}}. \quad (8)$$

*Says:* bond-implied CDS = (coupon-vs-Libor differential weighted by risky annuities) + (bond-discount term per unit risky annuity). The form makes each input's contribution legible.

**CDS-bond basis decomposition (paper's eq. 8):**

$$\text{Basis} = S^{\text{CDS}} - S^{\text{ASW}} = \underbrace{(\bar{L} - \bar{L}^{\text{Risky}})}_{\text{(i) curve-slope}} - \underbrace{C \left( 1 - \frac{PV01}{PV01^{\text{Risky}}} \right)}_{\text{(ii) accrual asymmetry, } \geq 0 \text{ so subtracted}} + \underbrace{D \left( \frac{1}{PV01^{\text{Risky}}} - \frac{1}{A} \right)}_{\text{(iii) off-par}}. \quad (9)$$

*Says:* three additive contributions to the basis:

- (i) sign of curve slope: positive when forward Libor is upward sloping, negative for inverted, zero for flat.
- (ii) bond/CDS accrual asymmetry: always negative contribution to the basis since  $PV01 < PV01^{\text{Risky}}$ ? Actually  $PV01^{\text{Risky}} < PV01$  when survival is below 1; the paper writes the term so the net contribution is negative. (Coupon accrued but unpaid on bond default; CDS premium accrued and *paid* on default.)
- (iii) off-par effect: positive for discount bonds ( $D > 0$ ), negative for premium bonds, vanishing at par.

**3 Derivations & commentary**

### Commentary

The cash-flow accounting that produces (7) (Appendix A in the paper) is worth internalising: a negative-basis trade has zero initial value, three streams of running cash-flows (net coupon, accrued premiums on default, terminal redemption on survival), and the no-arbitrage condition  $\mathbb{E}^Q[\cdot] = 0$  applied with the Lando-formula factorisation  $\mathbb{E}^Q[\mathbf{1}_{\tau > T} \cdot Z(T)] = DF(T) P(\tau > T)$  (under credit-rate independence) gives exactly (7). The trapezoidal approximation of the default-leg integral produces the  $(P_{k-1} + P_k)/2$  weights in Loss.

### Commentary

**Consistency theorem (Appendix B).** The proposition shows (7)  $\Rightarrow$  (5) under (6). The key algebraic identity is  $\text{Loss} = (1 - R) [\text{tail-recovery term}] + L \cdot \text{PV01} - L^{\text{Risky}} \cdot \text{PV01}^{\text{Risky}}$ , which lets the explicit form be rewritten in terms of  $(1 - R) \text{Loss} / \text{PV01}$ . The reverse implication does *not* hold: a CDS spread quoted in the market with an assumed flat recovery  $R$  may not satisfy (6) on the corresponding bond, which is the empirical CDS-bond basis.

### Commentary

The paper's three application scenarios are clean and worth lifting: **(2.3.1)** single bond: assume  $R$ , solve (6) for constant  $\lambda$ , plug into (8). **(2.3.2)** bond term structure: bootstrap a piecewise-constant  $\lambda$  across maturities. **(2.3.3)** both CDS and bond term structures: solve (8) for  $\lambda$  using market CDS spreads, then use (6) to solve for an implied recovery-rate term structure — comparable to recovery-swap quotes.

### Commentary

The interpretation of  $W$  in (3) is operationally important. A natural alternative is to buy CDS on full  $(1 - D)$  notional, but this leaves a residual exposure  $DR$  on default. The paper notes (Remark 1 after eq. 7) that the simpler  $1 - D$  notional gives the *same* formula (8) for the fair spread; the smaller default payoff is offset by larger carry on the loan. So the choice of CDS notional is presentational, not mathematical.

### Commentary

**Libor as “risk-free”.** The paper takes Libor as risk-free because at the time CDS markets discounted at Libor (this is 2008, pre-OIS regime). Modern practice would use OIS for both bond discounting and CDS pricing, and reinstate a separate funding-vs-discount distinction. The structural formulas (7)–(9) are unchanged under OIS discounting; only the input curve changes.

## 4 Validation

### Validation

This is a *specification* of mathematical objects the paper defines and the numerical values they should take. It says nothing about how those objects are implemented or invoked.

**Quantities the paper defines:**

- Risk-free annuity  $A$  and DF-weighted Libor  $\bar{L}$  (1).

- Par asset swap spread  $S^{\text{ASW}}$  via (2) (two equivalent forms).
- Default-neutral CDS notional ratio  $W$  (3).
- Risky annuity  $\text{PV01}^{\text{Risky}}$  and default-leg integral Loss (4).
- Bond-pricing consistency identity (6).
- Bond-implied CDS spread (7) and its simplified form (8).
- Standard CDS spread (5).
- CDS-bond basis decomposition (9).

**Canonical numerical case** (Tables 1 & 2):

- Bond: 10Y, 7% semiannual coupon, 30/360 day count.
- $R = 40\%$ .
- Libor: swap zero curve for 16-July-2008 from Bloomberg (3M = 2.8% to 10Y = 4.8%, linearly interpolated to payment dates). Table 2 uses flat 4.7% as a comparison.
- Calibration mode: single-bond (§2.3.1), constant  $\lambda$  from (6).
- $D \in \{-10\%, -5\%, 0\%, 5\%, 10\%, 15\%, 20\%\}$  (bond market price ranges from 110 to 80).

**Expected numerical values** (Table 1, market curve; all in %):

| $D$ (%)                                         | −10   | −5    | 0    | 5    | 10   | 15   | 20    |
|-------------------------------------------------|-------|-------|------|------|------|------|-------|
| $W$                                             | 1.06  | 1.03  | 1.00 | 0.96 | 0.93 | 0.88 | 0.83  |
| $S^{\text{CDS}}$                                | 0.96  | 1.60  | 2.31 | 3.09 | 3.97 | 4.97 | 6.13  |
| $S^{\text{ASW}}$                                | 1.05  | 1.67  | 2.29 | 2.92 | 3.54 | 4.16 | 4.79  |
| Basis                                           | −0.09 | −0.07 | 0.01 | 0.17 | 0.43 | 0.81 | 1.34  |
| $\lambda$                                       | 1.58  | 2.64  | 3.80 | 5.09 | 6.65 | 8.19 | 10.11 |
| $\bar{L} - \bar{L}^{\text{Risky}}$              | 0.03  | 0.05  | 0.08 | 0.11 | 0.14 | 0.17 | 0.21  |
| $C(1 - \text{PV01}/\text{PV01}^{\text{Risky}})$ | 0.03  | 0.05  | 0.07 | 0.09 | 0.11 | 0.14 | 0.18  |
| $D/\text{PV01}^{\text{Risky}} - D/A$            | −0.09 | −0.07 | 0.00 | 0.15 | 0.40 | 0.78 | 1.31  |

Table 2 (flat 4.7% Libor curve, same bond):

| $D$ (%)          | −10   | −5    | 0     | 5    | 10   | 15   | 20   |
|------------------|-------|-------|-------|------|------|------|------|
| $S^{\text{CDS}}$ | 0.92  | 1.55  | 2.24  | 3.00 | 3.86 | 4.83 | 5.95 |
| $S^{\text{ASW}}$ | 1.04  | 1.67  | 2.30  | 2.93 | 3.56 | 4.20 | 4.83 |
| Basis            | −0.12 | −0.12 | −0.06 | 0.07 | 0.29 | 0.63 | 1.12 |
| $\lambda$        | 1.51  | 2.55  | 3.68  | 4.94 | 6.35 | 7.96 | 9.81 |

**Equation  $\leftrightarrow$  quantity map:**

| Paper eq. | Symbol                                    | Quantity                                                       |
|-----------|-------------------------------------------|----------------------------------------------------------------|
| (1)       | $A, \bar{L}$                              | risk-free annuity, DF-weighted Libor                           |
| (2)       | $S^{\text{ASW}}$                          | par asset swap spread (two forms)                              |
| (3)       | $W$                                       | default-neutral CDS notional ratio                             |
| (4)       | $\text{PV01}^{\text{Risky}}, \text{Loss}$ | risky annuity, default-leg integral                            |
| (5)       | $S^{\text{CDS}}$ (standard)               | textbook CDS fair-spread identity                              |
| (6)       | bond consistency                          | bond price as coupon + recovery + survival-discounted notional |
| (7)–(8)   | $S^{\text{CDS}}$ (bond-implied)           | explicit and simplified forms                                  |
| (9)       | Basis                                     | 3-term decomposition of CDS-bond basis                         |

**Invariants and identities to verify:**

1. *Two ASW forms agree:*  $(M - (1 - D))/A = C - \bar{L} - D/A$  identically (given the definitions of  $M, A, \bar{L}$ ); the second form is the first rewritten using  $M = \sum C \Delta T_{k-1} DF(T_k) + DF(T_N) = C \cdot A + 1$  for a par bond on the risk-free curve.
2. *Default-neutral notional:* with CDS notional  $(1 - D)/(1 - R)$  and recovery  $R$ , the default payment is  $(1 - D)$ , exactly matching the loan repayment. So the combined position has zero default exposure by construction.
3. *Consistency theorem:* under (6), the explicit (7) equals the textbook (5).
4. *Reproduction of Table 1:* at the calibrated  $\lambda$  given in row 5,  $S^{\text{CDS}}, S^{\text{ASW}}, \text{Basis}$  in the upper rows should be reproduced exactly. The three basis-decomposition rows must sum (with signs as in (9)) to the Basis row above.
5. *Curve-slope  $\rightarrow$  basis sign:* in Table 2 (flat curve), the first basis term  $\bar{L} - \bar{L}^{\text{Risky}} = 0$  identically; the basis becomes (mostly) negative for low  $D$  and turns positive only as the off-par term dominates.
6. *Par bond basis  $\neq 0$ :* at  $D = 0$ , Basis = +0.01% on the upward-sloping curve and -0.06% on the flat curve — showing that even par bonds have non-zero basis due to the accrual-asymmetry term.
7. *Monotonicity:*  $S^{\text{CDS}}, S^{\text{ASW}}, \lambda, \text{Basis}$  are all increasing in  $D$  (Table 1).  $S^{\text{CDS}}$  grows faster than  $S^{\text{ASW}}$ , which is why Basis widens for deep-discount bonds.
8. *W monotonicity in D:*  $W$  decreases in  $D$  for fixed  $R$ ; at  $D = 0$ ,  $W = 1$ . Check:  $W = (1 - D/(1 - R))/(1 - D)$ , differentiated and shown to be negative for  $R < 1$ .

## 5 Glossary of symbols (paper's notation)

|                          |                                                         |
|--------------------------|---------------------------------------------------------|
| $T_0, T_k$               | Today; bond / ASW payment dates                         |
| $\Delta T_k$             | Year fraction of period $(T_{k-1}, T_k)$                |
| $C$                      | Bond fixed coupon (paper examples: 7% semiannual)       |
| $M$                      | Bond risk-free price                                    |
| $1 - D$                  | Bond market dirty price; $D$ is the discount            |
| $R$                      | Expected recovery rate                                  |
| $W$                      | CDS-notional-to-bond-price ratio for default neutrality |
| $L_{k-1}$                | Forward Libor for period $(T_{k-1}, T_k)$ at $T_0$      |
| $DF(T_k)$                | Risk-free discount factor at $T_0$                      |
| $P(\tau > T_k)$          | Survival probability from $T_0$ to $T_k$                |
| $\tau$                   | Default time                                            |
| $\lambda$                | Hazard rate (constant in §2.3.1)                        |
| $A$                      | Risk-free annuity                                       |
| PV01                     | Risky annuity (uses $DF$ and $P$ )                      |
| PV01 <sup>Risky</sup>    | Survival-weighted risky annuity (paper's notation)      |
| $\bar{L}$                | $A$ -weighted average forward Libor                     |
| $\bar{L}^{\text{Risky}}$ | Risky-annuity-weighted average forward Libor            |
| Loss                     | Risky default-leg integral, trapezoidal-approximated    |
| $S^{\text{ASW}}$         | Par asset swap spread                                   |
| $S^{\text{CDS}}$         | Bond-implied CDS spread                                 |
| Basis                    | $S^{\text{CDS}} - S^{\text{ASW}}$ , CDS-bond basis      |

### TODO / Open questions

- Reproduce Table 1 and Table 2 on the 10Y, 7%-coupon bond with  $R = 40\%$ , using the 16-July-2008 Bloomberg swap zero curve. Match across the seven  $D$  levels.
- Verify the 3-term basis decomposition sums to the Basis row in both tables. The

third row (off-par term) is given as  $D/PV01^{\text{Risky}} - D/A$ , which differs by a sign convention from (9); check.

- Implement §2.3.2 bootstrap: given a bond price term structure on a single issuer, recover a piecewise-constant  $\lambda$  from (6).
- Implement §2.3.3: given both market CDS spreads and bond prices on the same issuer, jointly solve for  $\lambda(T)$  and  $R(T)$ . Compare implied  $R(T)$  to any available recovery swap quotes.
- Modern (OIS) update: re-do Table 1 with OIS discounting in place of Libor; quantify the impact on  $S^{\text{CDS}}$ ,  $S^{\text{ASW}}$ , and Basis.
- Test the par-bond basis identity at  $D = 0$  across various curve shapes (flat, upward, inverted, humped) to verify the sign predictions of the three-term decomposition.